

Distilling Native Query–Key Responses into Compact Memory Landmarks

Jorge Simão

EInnovator R&D – Center for the Sciences of Computation, Cognition, and Complexity

August 2026

Abstract

Progressive Retrieval Attention (PRA) uses compact routing gists to choose long-memory chunks, then materializes the selected chunks’ unchanged native key/value (K/V) states. Mean gists are inexpensive but erase sparse token-level address signals. We test a narrower alternative: preserve the frozen transformer’s full key-stream response rather than reconstructing key vectors, clustering tokens, or generating text summaries. A full-QK teacher scores each 32-token chunk from all native keys. Controllers retain one to eight actual keys: last-token, random, farthest-first, a query-aware greedy oracle, or a 321-parameter key-only selector trained from oracle labels. Validation uses 16 identity-disjoint HotpotQA/QASPER questions; testing uses the frozen 32-question Paper 2.5/2.6 cohort, a fixed four-chunk materialization budget, and five selector seeds. The selected top-four-mean teacher passes a controlled sanity gate and improves natural recall over a mean key gist by 0.0004 on HotpotQA and 0.0276 on QASPER. Greedy landmarks expose headroom: at four keys they improve QASPER recall by 0.0802, although its paired 95% interval $[-0.0142, 0.2233]$ includes zero. The learned eight-key selector improves QASPER recall by 0.0413 over the mean-key control and by 0.1010 over Paper 2.6’s semantic gist baseline; only the latter interval excludes zero, narrowly. It is unchanged on HotpotQA, varies substantially across seeds, and recovers 40.5% of the oracle top-four-overlap gain, below the prespecified 80% gate. We therefore stop before synthetic slots or streaming memory. The result identifies both an opportunity and a boundary: native response geometry contains useful QASPER signal, but a tiny query-independent token selector does not reliably distill it across routing regimes.

1 Introduction

PRA separates *routing* from *reading*. A small summary chooses memory chunks; only the chosen chunks contribute their original token K/V to attention. The separation preserves pretrained values and bounds active memory, but it makes routing-summary quality consequential. One mean vector per chunk is attractive because it is deterministic, cacheable, and cheap. It is also a severe compression: a sparse key that responds strongly to a future query can disappear inside 31 unrelated tokens.

Earlier work already tested natural repairs. Multi-gist, prototype, k -means, SOM, local-gist, native-QK closure, lexical, and graph-query variants sometimes helped a dataset or controlled fixture, but did not establish a uniform natural frontier. Those methods mainly organize semantic or local geometry. This paper asks a different question: can a compact native-key set preserve the *response surface* that the frozen attention layer assigns to the full chunk?

The distinction matters. A key is useful because of how plausible queries address it, not because it reconstructs the mean of its neighbors or reads well as text. We therefore use full token-level keys as a teacher, measure ranking preservation directly, and keep the downstream materialization budget fixed. The study is gated. We first verify inherited Paper 2.6 rows, then test whether the teacher is

meaningful, then ask whether a query-aware oracle has headroom. A learned selector is permitted only after those checks; more expressive off-manifold slots and recurrent memory remain closed if it fails.

The contributions are:

1. a tensor-native definition of chunk response distillation under grouped-query attention, with max, top- r , log-sum-exp, and attention-mass teachers;
2. matched native-key controllers, response metrics, routing metrics, changed-selection audits, and synchronized CUDA cost measurements;
3. a controlled 32–256-token suite and a frozen natural evaluation aligned with the Paper 2.5/2.6 32-token chunk and four-chunk budget;
4. a measured boundary: oracle landmarks show natural QASPER headroom, while a 321-parameter key-only selector recovers too little of that headroom to pass deployment gates.

2 Native response distillation

2.1 Full-key teacher

Let chunk C contain pre-RoPE native keys $K_C \in \mathbb{R}^{n \times H_{kv} \times d_h}$ and let a frozen decoder produce query heads $q \in \mathbb{R}^{H_q \times d_h}$. The grouped-query map $g(h)$ assigns each query head to its native key head. Scaled token responses are

$$a_{i,h}(q, C) = \frac{q_h^\top k_{i,g(h)}}{\sqrt{d_h}}. \quad (1)$$

The teacher reduces tokens per head and then averages heads:

$$S^*(q, C) = \frac{1}{H_q} \sum_h F(a_{1,h}, \dots, a_{n,h}). \quad (2)$$

We validate four reductions: maximum, top- r mean, length-normalized log-sum-exp, and a softmax attention-mass proxy. Validation selects top-four mean with $r = 4$. This selection occurs before test evaluation and is frozen for every controller.

For compact landmarks $G(C) = \{\hat{k}_1, \dots, \hat{k}_m\}$, the estimate $\hat{S}(q, C)$ applies the same F and grouped-head mapping. The primary target is not $\|K_C - \hat{K}_C\|$. It is agreement between S^* and $\hat{S}$ over candidate chunks: score error, rank correlation, teacher top- k overlap, and KL divergence between softmax-normalized chunk distributions.

2.2 Controllers

All deployable landmark outputs are actual pretrained keys. Mean is the only synthetic control.

Mean Average valid keys independently in each native key head, yielding one key per chunk.

Last and random

Retain the final m valid keys or a deterministic random subset.

Farthest-first

Flatten native heads per token and greedily retain cosine-diverse keys.

Greedy oracle

Add the token that most reduces full-response squared error. It sees the evaluation query and is explicitly non-deployable.

Learned selector

Score each token from overall key norm, mean head norm, head-norm variation, cosine to the chunk centroid, local key change, forward/reverse position, and validity. A two-layer MLP, $8 \rightarrow 32 \rightarrow 1$, has 321 parameters. Hard top- m is used at inference.

Oracle selections for $m = 8$ supervise the learned model. Five seeds train for 400 AdamW steps on 16 validation identities. The backbone, native projections, tokenizer, and all natural test identities remain frozen. The selector receives no question vector, evidence label, dataset identifier, lexical feature, or chunk-level answer supervision at inference.

2.3 Matched materialization

The intervention changes only routing summaries. Candidate units are the same 32-token local chunks. Every method requests four chunks, so it materializes approximately 128 native K/V tokens: 126.1 on HotpotQA because of short final chunks and 128.0 on QASPER. Active memory fractions are 8.8% and 3.3%, respectively. Values are never compressed, projected, or synthesized.

3 Protocol

3.1 Frozen natural cohort

We reuse Qwen3-0.6B revision `c1899de2`, layer 27, 16 query heads, eight K/V heads, and head width 128. Source blocks are encoded in bounded 256-token calls, then divided into 32-token routing chunks without local re-encoding. Pre-RoPE keys and attention-input query states preserve token spans and evidence masks. The query state is projected through the unchanged native `q_proj` and `q_norm`.

Validation contains eight HotpotQA and eight QASPER identities. Test contains 16 of each, matching the frozen Paper 2.5/2.6 identities and excluding validation identities. Feature tensors are 329 MB and 621 MB, respectively; manifests record SHA-256 hashes. Paper 2.6’s 224 test rows are re-aggregated exactly at G0. Paper 2.5’s 256-token-parent, 20%-budget rows are retained only as historical context because their candidate unit and budget are not numerically matched.

3.2 Controlled suite

Twenty synthetic cases cross chunk sizes $n \in \{32, 64, 128, 256\}$. Each case has 16 candidate chunks. An evidence chunk contains sparse query-aligned native keys while a distractor has a coherent but weaker mean. This tests whether full-key and landmark controllers preserve a sparse address that averaging can hide. The suite is a mechanism check, not evidence for natural language performance.

3.3 Metrics and uncertainty

Native preservation metrics are MAE, RMSE, Spearman ρ , Kendall τ , KL divergence, and teacher top- k overlap for $k \in \{1, 2, 4, 8\}$. Routing metrics are evidence recall and precision, any-evidence recall, contiguous evidence-group completion, exact selected identity, MRR, active memory fraction, evidence-normalized overhead, materialized tokens, native dots, and wall time. Five-thousand-draw paired bootstrap intervals resample identities. Selector seeds are averaged within identity before comparisons with deterministic baselines.

The prespecified gates are:

- G0: inherited Paper 2.6 aggregation parity;
- G1: full-K beats mean on controlled recall and at least one natural dataset;

Table 1: Held-out four-chunk routing. Learned rows average five seeds. Oracle rows use the test query and are non-deployable. Values are evidence recall; parentheses give teacher top-four overlap.

Method	HotpotQA	QASPER
Mean key, $m = 1$	0.124 (0.594)	0.098 (0.516)
Full-K teacher	0.124 (1.000)	0.125 (1.000)
Farthest-first, $m = 2$	0.142 (0.734)	0.130 (0.422)
Greedy oracle, $m = 4$	0.145 (0.922)	0.178 (0.703)
Greedy oracle, $m = 8$	0.124 (0.984)	0.136 (0.812)
Learned selector, $m = 4$	0.122 (0.741)	0.083 (0.422)
Learned selector, $m = 8$	0.124 (0.834)	0.139 (0.553)

- G2: greedy $m \leq 8$ improves teacher top-four overlap and at least one natural recall result without changing materialized K/V;
- G3: the learned selector recovers at least 80% of oracle gain, improves recall or completion on one natural dataset, and regresses by less than 0.02 elsewhere;
- G4/G5: synthetic slots and streaming memory run only after G3 passes.

4 Results

4.1 Teacher sanity and oracle headroom

G0 passes with zero mismatches across 224 inherited rows. Validation selects top-four mean across query heads: its validation evidence recall is 0.1387 versus 0.1255 for the corresponding mean-key control. Max, log-sum-exp, and attention-mass teachers do not beat that control on validation.

G1 also passes, narrowly on natural data. Full-K recall is 1.000 versus 0.950 for mean in the controlled suite. On held-out HotpotQA it is 0.1240 versus 0.1235; on QASPER it is 0.1252 versus 0.0976. The natural teacher is therefore not a strong retrieval oracle by itself. It provides a modest QASPER signal and essentially no Hotpot gain.

G2 passes. Relative to mean, greedy $m = 8$ improves aggregate teacher top-four overlap by 0.3438. Recall is not monotone in m : $m = 4$ is best, improving HotpotQA by 0.0219 and QASPER by 0.0802. The paired intervals, $[-0.0313, 0.0750]$ and $[-0.0142, 0.2233]$, both include zero. Oracle landmarks therefore establish response-preservation headroom and a promising QASPER direction, not a statistically resolved natural retrieval gain.

4.2 Learned selector: useful direction, failed gate

The learned selector does not recover the oracle. At $m = 8$, it captures 40.5% of the oracle teacher-overlap gain, below the 80% threshold. Its HotpotQA recall delta over the QK mean is +0.0004 with interval $[-0.0573, 0.0533]$. Its QASPER delta is +0.0413 with interval $[-0.0589, 0.1670]$. G3 therefore fails and G4/G5 do not run.

Seed variation is material. HotpotQA $m = 8$ recall ranges from 0.1173 to 0.1298. QASPER ranges from 0.0478 to 0.1821; four seeds are at or above 0.119, while seed 37 collapses. Teacher top-four overlap is more stable than evidence recall, showing that response imitation and evidence utility are related but not interchangeable.

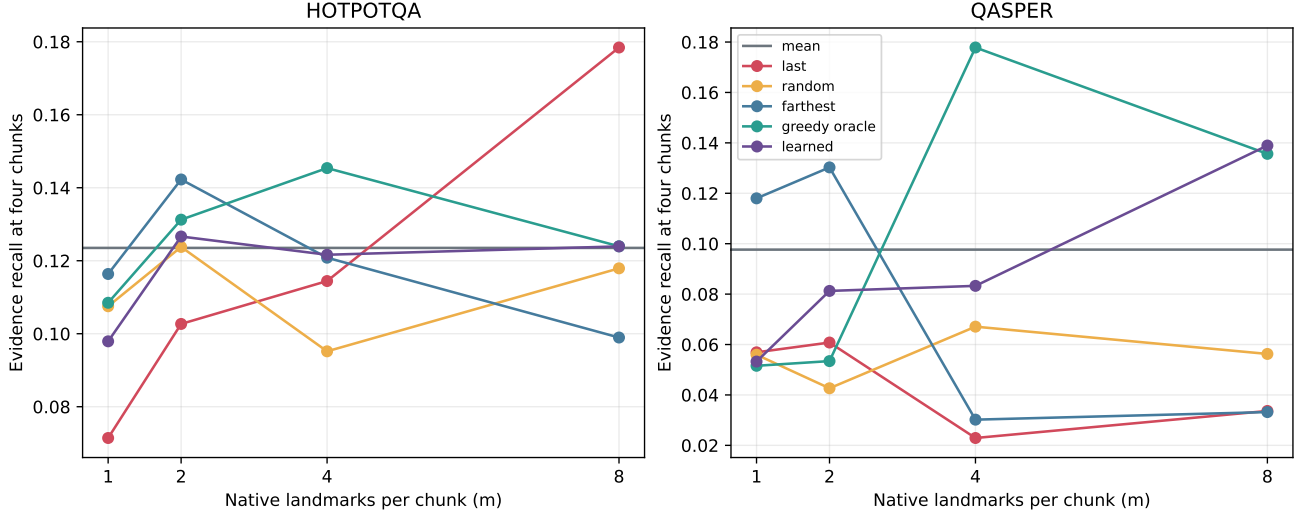


Figure 1: Natural evidence recall under the matched four-chunk materialization budget. The dataset-separated view exposes the Hotpot/QASPER asymmetry.

4.3 Comparison with Paper 2.6

Figure 2 compares the same frozen identities and four-chunk budget. Learned QK $m = 8$ is not a new overall frontier. On HotpotQA its 0.124 recall is below Paper 2.6 BM25 (0.398), exact (0.287), and iterative hybrid (0.279), with paired intervals excluding zero against all three. On QASPER it reaches 0.139: above Paper 2.6’s semantic gist (0.038) by 0.1010, with paired interval $[0.00004, 0.2270]$; above iterative hybrid (0.081) by 0.0579, interval $[-0.0610, 0.1974]$; and below exact routing (0.178) by 0.0387, interval $[-0.1951, 0.1334]$. The positive, defensible claim is dataset-specific: native response distillation improves the old QASPER semantic-gist baseline, but does not displace lexical exact matching and does not transfer to Hotpot.

Paper 2.5’s native-QK closure used 256-token parents and a 20% final-parent budget. At that setting, its top-four native closure reached chain completion 0.125 on HotpotQA and 0.875 on QASPER, compared with one-shot parent values 0.138 and 0.938. Those rows concern associative successor discovery, not one-shot 32-token ranking. We preserve them as historical controls but do not pool their metrics with Table 1.

4.4 Cost

The synchronized CUDA audit includes uncached controller construction, compact-key gathering, and QK scoring. Table 2 reports means over all 32 test identities on an NVIDIA GTX 950M. Absolute times are hardware-specific; the ordering and native-dot counts expose algorithmic differences.

The learned selector is 4.5–5.0 times faster than farthest-first and about 237–271 times faster than the top- r greedy oracle, but 16.9–24.4 times slower than mean on this unoptimized Python/CUDA path. Greedy cost is dominated by query-aware candidate search, not compressed scoring. The quality CSV’s `compression_ms` field times compact assembly after indices are cached; Table 2 instead uses the separate synchronized uncached audit.

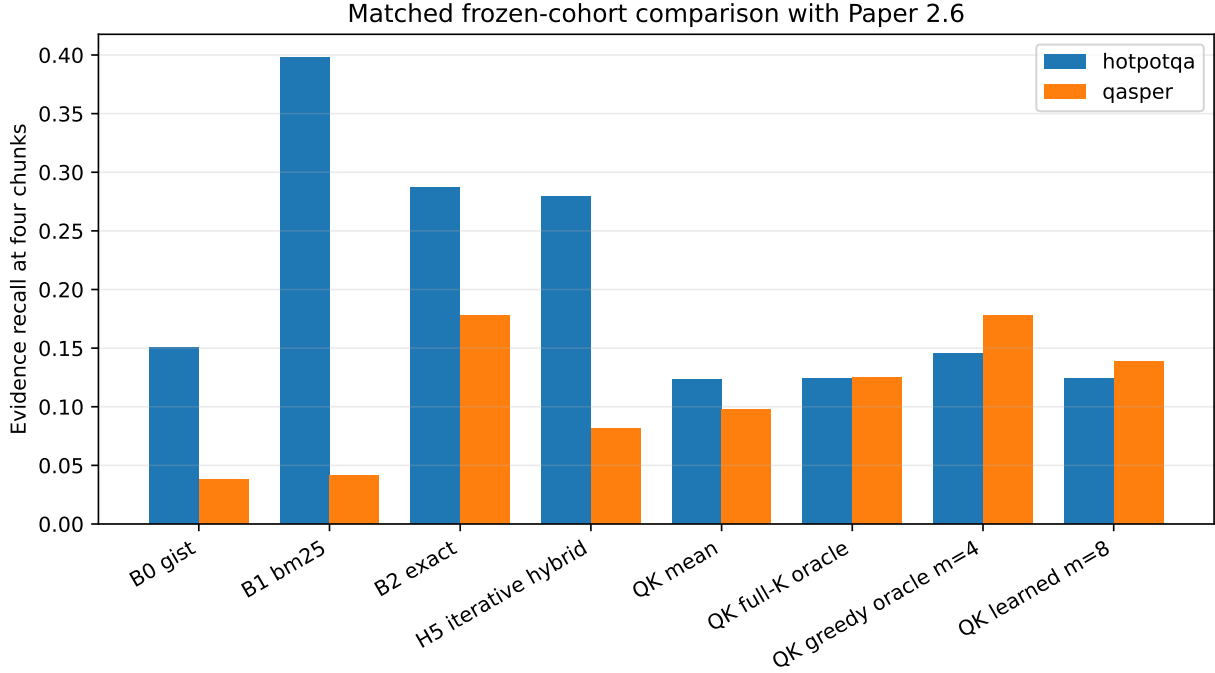


Figure 2: Matched Paper 2.6 comparison. Full-K and greedy rows are non-deployable upper bounds. Learned QK is competitive on QASPER but not Hotpot.

Table 2: End-to-end routing-controller cost at $m = 8$. Mean uses $m = 1$. The full-K row performs no selection.

Method	HotpotQA (50.1 chunks)		QASPER (142.6 chunks)	
	ms	native dots	ms	native dots
Mean	6.1	802	10.7	2,282
Full-K	25.9	24,898	34.1	70,853
Farthest-first	463.1	6,416	1,308.4	18,256
Greedy oracle	24,367.6	6,416	70,376.0	18,256
Learned selector	102.8	6,416	259.9	18,256

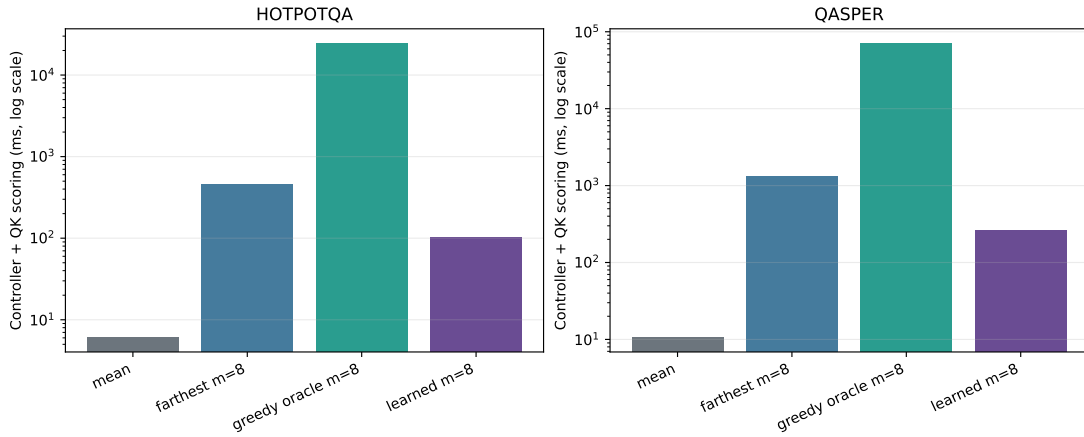


Figure 3: Controller construction plus QK scoring on a log scale. The oracle is a feasibility instrument, not a serving method.

4.5 Gate summary

Table 3: Prespecified decisions. The sequence stops at G3.

Gate	Status	Evidence
G0	Pass	224 inherited Paper 2.6 rows reproduce with zero aggregate mismatches.
G1	Pass	Controlled recall 1.00 vs. 0.95; full-K improves QASPER mean-key recall by 0.0276.
G2	Pass	Greedy $m = 8$ top-four overlap +0.3438; greedy $m = 4$ QASPER recall +0.0802.
G3	Fail	Learned selector recovers 40.5% of oracle gain, below 80%; natural intervals include zero.
G4	Closed	Synthetic off-manifold slots require G3.
G5	Closed	Streaming recurrent memory requires an offline compressor pass.

5 What the result means

Three observations explain the outcome.

First, a full-key teacher is not automatically an evidence oracle. Native attention keys encode many functions besides document retrieval, and a question’s final layer-27 query can respond to syntactic, discourse, and answer-format features. Top-four mean is the best validated reduction here, but its natural advantage over a mean key is small.

Second, response approximation and top-four selection are different objectives. Greedy $m = 8$ nearly reproduces the teacher ranking on HotpotQA yet does not improve recall. On QASPER, $m = 4$ retrieves more evidence than $m = 8$ despite lower teacher overlap. Additional landmarks can faithfully restore teacher preferences that are irrelevant or harmful to evidence selection. A future objective should combine teacher distillation with validation-only ranking calibration rather than treating the teacher as ground truth.

Third, the key-only selector cannot condition on the current information need. It learns generic token salience from norm, novelty, position, and local change. That is enough to improve the old QASPER semantic baseline, where sparse technical terms and localized evidence are useful, but not enough for Hotpot’s cross-document relational routing. The oracle/learned gap is therefore plausibly a missing conditioning signal, not merely insufficient MLP width.

The next justified experiment is a query-conditioned but still bounded selector, for example a low-rank bilinear score between a cached token feature and the current native query, trained with listwise chunk-ranking loss. It should be evaluated against QASPER exact routing and Hotpot BM25/hybrid, not only against a mean gist. Synthetic slots and recurrent state are premature until a native-key selector closes more of the oracle gap.

6 Relation to prior PRA results

Paper 1 established mean-gist routing and selected native K/V. Paper 2.5 tested iterative associative closure and showed that local/native propagation can solve controlled bridges without yielding a uniform natural gain. Paper 2.6 demonstrated strong channel asymmetry: lexical evidence is particularly effective for Hotpot and exact matching for QASPER. Paper 2.7 moved structure discovery to the query side and found that better natural facet recovery could still reduce retrieval.

Paper 2.8 does not rename those mechanisms. It changes the memory-summary objective. Its negative Hotpot result agrees with Paper 2.6: lexical relational discovery dominates a native response summary. Its QASPER result adds something narrower: an eight-key response-distilled controller can outperform the original semantic gist and approach, but not exceed, exact matching under the

same four-chunk budget. Query-graph facets remain a query-side control and are not used to train the memory selector.

7 Limitations

The natural cohort has 32 test identities and one frozen model, layer, tokenizer, chunk size, and materialization budget. Five seeds quantify selector initialization, not dataset sampling uncertainty. The QASPER gain over the Paper 2.6 semantic baseline is barely separated from zero and should be replicated on a larger identity-disjoint cohort. Hotpot and QASPER do not cover the 2Wiki/MuSiQue settings proposed in the initial plan because G3 failed before expansion. The controlled suite contains only 20 cases and is deliberately schematic.

The teacher is selected from five predefined reductions but does not include post-RoPE distance, value utility, downstream answer log-probability, or layer combinations. Oracle landmarks see the test query and cannot be deployed. Learned targets are generated by that expensive oracle and inherit its errors. The learned selector is query-independent and uses only eight hand-designed key features. Runtime is measured with eager Python/PyTorch on an old mobile GPU; optimized batched kernels would change absolute latency. No native K/V answer generation is run because the routing-quality gate fails.

8 Conclusion

Compact native keys can preserve more of a frozen transformer’s chunk-response geometry than one mean key, and that geometry contains useful QASPER routing signal. The result is not a general learned-compressor victory. Full-K itself is a weak natural evidence teacher; greedy landmarks show uncertain oracle headroom; and a tiny key-only selector captures 40.5% of the preservation gain, varies across seeds, loses to strong Hotpot channels, and does not beat QASPER exact matching. The gate sequence therefore stops before synthetic slots and streaming memory. The clearest next step is not more unconditional capacity, but bounded query-conditioned selection with an evidence-calibrated ranking objective.

A Reimplementation map

The implementation separates mechanism from experiment policy.

- `src/prahf/qk_compression.py` defines GQA head mapping, full-key teachers, landmark gathering, no-training selectors, the 321-parameter selector, response metrics, and routing metrics.
- The Paper 2.5 feature precomputer, `precompute_native_qk_features.py`, now supports identity-disjoint validation/test offsets while retaining the historical default.
- `experiments/paper2.8_qk_compression/run_gated_study.py` selects the teacher on validation, runs gates in order, trains five seeds only after G2, and emits row-level artifacts.
- `run_cost_benchmark.py` times uncached controller construction separately from cached quality evaluation; `summarize_results.py` builds matched Paper 2.5/2.6 tables and plots.

The core serving path is:

```
q = native_q_projection(frozen_query_state)    # [Hq, Dh]
K = cached_pre_rope_keys                       # [C, T, Hkv, Dh]
```



```

x = key_only_features(K) # [C, T, 8]
z = selector(x).masked_fill(~token_mask, -inf) # [C, T]
landmarks = gather(K, topk(z, m=8)) # [C, 8, Hkv, Dh]
scores = top4_mean(q @ landmarks / sqrt(Dh)) # [C]
chunks = stable_topk(scores, k=4)
materialize_native_kv(chunks) # unchanged 32-token K/V

```

The oracle substitutes a query-aware greedy search for `selector(x)` and must not be used online.

B Artifacts

Tracked results live under `docs/papers/shared/results/paper2_8_qk_compression/`. The directory contains natural and synthetic row CSVs, teacher validation, paired effects, changed-selection audits, matched Paper 2.6 and historical Paper 2.5 controls, five small selector checkpoints, synchronized cost rows, plots, gate decisions, and manifests. The 329 MB validation and 621 MB test Q/K tensors are ignored; manifests provide their hashes and regeneration commands. The full test cache hash is `edc589cc...75993ae8`; validation is `87b85b25...7533cf8`.

References

- [1] Simão, J. Progressive Retrieval Attention: Model-Bounded Sparse Native-KV for Long Context and URI-Addressed Memory. Project manuscript, 2026.
- [2] Simão, J. Iterative PRA and Associative Memory Closure. Project manuscript, 2026.
- [3] Simão, J. Hybrid Lexical-Semantic Routing for Progressive Retrieval Attention. Project manuscript, 2026.
- [4] Simão, J. Latent Query Decomposition from Frozen Transformer State Graphs. Project manuscript, 2026.
- [5] Yang, Z. et al. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. EMNLP, 2018.
- [6] Dasigi, P. et al. A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers. NAACL, 2021.